

Lab 5 addresses practical Tool Center Point (TCP) calibration for the *myCobot 280*. The focus is the geometric offset between the robot flange and the actual tool point, and how this offset influences Cartesian accuracy in a real robot workflow.

The lab is solved with Python in Jupyter notebooks and is based on measurements collected on the physical robot. Students work with the provided calibration workflow, estimate a flange-to-TCP offset from recorded poses, and assess the quality of the result before using the calibrated TCP in later robot motions. Before starting the practical work, inspect the provided calibration workflow in the codebase carefully.

Learning Objectives TCP Calibration Lab

After completing this lab, students will understand why a correct TCP model is required for meaningful Cartesian robot motion with the *myCobot 280*. They will be able to distinguish the robot flange frame from the actual TCP frame of the mounted tool, collect suitable measurements on the real robot, and estimate the flange-to-TCP offset from these measurements.

By the end of the lab, students will be able to:

- Explain the role of the TCP in practical robot programming and execution.
- Interpret the flange-to-TCP offset as a rigid geometric parameter of the mounted tool.
- Record and organize robot measurements for TCP calibration on the *myCobot 280*.
- Estimate a TCP offset from multiple measured robot poses.
- Validate a calibration result using residual, reprojection, or Cartesian position error measures.
- Apply the calibrated TCP in the practical robot workflow.

Short Questions

- 1) What is a Tool Center Point (TCP) in robot manipulation, and why is it usually not identical to the robot flange origin?
- 2) Why is TCP calibration necessary before executing Cartesian motions with a real robot system such as the *myCobot 280*?
- 3) Explain the difference between the flange frame and the TCP frame of a mounted tool.
- 4) Which measurements are required to estimate a flange-to-TCP offset from repeated robot poses?
- 5) What is the basic idea behind estimating one TCP offset from several measurements of the real robot?
- 6) How can the quality of a TCP calibration be validated? Name suitable error measures and explain what they indicate.
- 7) What effect does an incorrect TCP have on positioning accuracy, contact motions, or repeated approach motions?

- 8) Why is it preferable to use several measurement poses instead of only one or two poses for TCP calibration?

TCP Calibration

For practical robot operation, it is not sufficient to describe only the kinematic chain up to the robot flange. Real manipulation tasks are performed with a mounted tool, and Cartesian commands must refer to the point on that tool that is physically relevant for the task. This point is the *Tool Center Point* (TCP). In the *myCobot 280* workflow, a correct TCP description is essential whenever the robot is expected to approach a target position accurately, touch a surface at a defined point, or repeat a motion with a consistent tool tip location.

What the TCP Is

The TCP is the reference point attached to the tool that represents the effective working point of the robot. Depending on the tool, it may be the center of a gripper, the tip of a pen, the end of a probe, or another physically meaningful point. Robot software uses this point as the Cartesian reference for motion planning, pose reporting, and target specification.

Mathematically, the TCP is described by a rigid transformation from the flange frame to the TCP frame. In this lab, the main quantity of interest is the translational offset from the flange origin to the tool point. If the flange frame is denoted by F and the TCP frame by T , then the flange-to-TCP transform can be written as

$${}^F\mathbf{T}_T = \begin{bmatrix} {}^F\mathbf{R}_T & {}^F\mathbf{t}_T \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (1)$$

where ${}^F\mathbf{t}_T$ is the TCP offset to be identified from measurements. In many practical calibration workflows, the dominant unknown is this offset vector.

Why TCP Calibration Matters

If the TCP is modeled incorrectly, a Cartesian target for the tool tip is not reached by the real tool tip. Even if the robot joints move exactly as commanded, the physical contact point of the tool is displaced because the geometric relation between flange and tool has been specified incorrectly. This leads to systematic positioning errors, poor repeatability in approach motions, and incorrect contact locations during practical use.

For the *myCobot 280*, TCP calibration therefore serves two purposes. First, it improves the geometric consistency between the robot model and the mounted tool. Second, it provides a practical basis for later workflows in which the calibrated TCP is used directly in Cartesian commands, recorded poses, or validation steps in the notebook.

Flange Frame and TCP Frame

The flange frame is the coordinate frame mechanically attached to the last robot link. It is fixed by the robot geometry and is available from the forward kinematics of the

manipulator itself. The TCP frame, by contrast, is attached to the mounted tool and depends on how the tool is installed.

These two frames are generally different. Their origins need not coincide, and their orientations may also differ. In this lab, the central calibration quantity is the offset from the flange origin to the actual tool point. Distinguishing these two frames is essential: the flange belongs to the robot, whereas the TCP belongs to the tool configuration currently mounted on the robot.

Measurement Idea and Calibration Setup

The practical idea of TCP calibration is to move the robot into several distinct poses while the physical tool point refers to the same effective spatial point, or while a repeated measurement relation is created from the real robot setup. Each measurement provides a flange pose from the robot model together with the known measurement condition from the experiment. Across several poses, these measurements constrain one common TCP offset.

In a typical measurement workflow, students command the *myCobot 280* to a sequence of robot poses and record the corresponding flange poses from the calibration notebook. The measurement poses should be sufficiently different in orientation and position so that the calibration is well-conditioned. Repeating nearly identical poses is usually not helpful, because it provides only weak additional information about the unknown offset.

The practical notebook contains the actual hands-on measurement steps and data handling. The purpose of this text is to explain the calibration principle and the interpretation of the result.

Estimating the Flange-to-TCP Offset

Assume that for measurement i the flange pose with respect to the robot base is known as

$${}^B\mathbf{T}_{F,i} = \begin{bmatrix} {}^B\mathbf{R}_{F,i} & {}^B\mathbf{p}_{F,i} \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (2)$$

If all measurements refer to the same physical TCP point in the workspace, then the corresponding TCP position in the base frame should ideally be identical for every pose:

$${}^B\mathbf{p}_{\text{TCP},i} = {}^B\mathbf{p}_{F,i} + {}^B\mathbf{R}_{F,i} {}^F\mathbf{t}_T \approx \mathbf{p}^*, \quad (3)$$

where ${}^F\mathbf{t}_T$ is the unknown flange-to-TCP offset and $\mathbf{p}^*$ is the common spatial point implied by the measurement setup.

Because real measurements are affected by noise, compliance, reading inaccuracies, and limited repeatability, the equations are not satisfied exactly. The calibration problem is

therefore formulated as an estimation problem. The unknown TCP offset is chosen such that the inconsistency across all measurements becomes as small as possible, typically by a least-squares criterion. In practice, this means finding the offset vector that best explains all recorded poses simultaneously rather than fitting one measurement at a time.

Using multiple measurements is important. It reduces the influence of noise, makes the estimate more robust, and allows the quality of the result to be checked quantitatively. The estimated TCP offset is only meaningful if it remains consistent across a sufficiently varied set of robot poses.

Calibration Implementation in the myCobot Codebase

In the practical workflow, the calibration procedure is provided by the script `fk_tool_calib.py`. Each measurement sample consists of the recorded joint angles in degrees, the measured TCP position in millimeters, and the measured TCP orientation given as roll-pitch-yaw angles in degrees. These samples are the direct input to the calibration computation.

For each sample, the script evaluates the forward kinematics from the joint angles to obtain the flange pose ${}^B\mathbf{T}_{F,i}$ with respect to the robot base. The measured TCP position and orientation are assembled into a homogeneous transformation ${}^B\mathbf{T}_{TCP,i}$ in the same base frame. The per-sample flange-to-TCP offset is then obtained by

$${}^F\mathbf{T}_{T,i} = \left({}^B\mathbf{T}_{F,i}\right)^{-1} {}^B\mathbf{T}_{TCP,i}, \quad (4)$$

which isolates the rigid transformation between the flange frame and the TCP frame for the current sample.

Averaging across all samples yields a single flange-to-TCP transform. The translational part is averaged component-wise, while the rotational part is averaged through a singular value decomposition based mean rotation, which guarantees that the resulting matrix is a valid element of $SO(3)$. In addition to the tool offset, the script also estimates a base correction transform, which compensates for a residual mismatch between the nominal robot base frame and the frame in which the measurements were taken.

Validating the Calibration

After estimating the TCP offset, the result must be validated before it is used in later robot operation. A useful calibration is not characterized only by the existence of a numerical solution, but by how well that solution explains the measured data.

One common validation idea is to recompute the TCP position for every recorded flange pose using the estimated offset and then compare these reconstructed points with the expected common point. The remaining differences are the residual errors of the calibration. Small residuals indicate that the estimated offset is geometrically consistent with the measurements.

Depending on the workflow, the calibration quality can be reported with error measures such as the mean residual norm, the maximum residual, or a root-mean-square position error. If image-based or projected measurements are involved, a reprojection-style error measure may also be considered. In all cases, the interpretation is the same: the smaller the remaining error after calibration, the better the estimated TCP explains the observed measurements.

Validation should also include a practical check on the robot. After applying the calibrated TCP in the notebook or robot interface, repeated approach motions toward the same target should become more consistent. This practical verification is important because it reveals whether the estimated TCP improves the actual robot workflow rather than only the numerical fit.

Using the Calibrated TCP in the Robot Workflow

Once the TCP offset has been estimated and validated, it becomes part of the tool description used in the practical workflow. Cartesian commands, reported end-effector poses, and target checks should then refer to the calibrated TCP rather than to the flange origin. This is the operational purpose of the entire procedure: the robot model should represent the real working point of the mounted tool as closely as possible.

For the *myCobot 280*, this means that subsequent notebook steps should use the calibrated TCP consistently whenever the tool position is interpreted in Cartesian space. A correct TCP does not change the robot mechanism itself, but it changes how the end-effector pose is computed and therefore directly affects practical positioning accuracy. Concretely, the estimated tool offset is applied through the launch configuration parameters `ik_tool_xyz` and `ik_tool_rpy`, and is used by the inverse kinematics solver via the inverse of the tool transform so that Cartesian targets are interpreted at the calibrated TCP rather than at the flange.

Competences

After successfully completing the TCP Calibration lab, students will have acquired the following competences:

- They understand the practical meaning of the TCP for Cartesian robot motion and accuracy.
- They can distinguish flange frame and TCP frame for a mounted tool on the *myCobot 280*.
- They are able to collect and interpret measurement data for TCP calibration.
- They can estimate and validate a flange-to-TCP offset from repeated robot measurements.
- They can use a calibrated TCP consistently in the subsequent robot workflow.